Customer Lifetime Value Analysis for Oberalp Sales Data

Abhishek Bargujar & Burak Toptaş

Supervisor: Floriano Zini

Free University of Bolzano

Introduction

Customer Lifetime Value is a crucial metric in business analytics, helping companies predict the total revenue a customer will generate over their relationship with the business. Understanding and forecasting CLV enables companies to make informed decisions regarding marketing budgets, customer retention strategies, and overall business growth.

Problem Statement

The primary problem addressed in this project is how to accurately estimate and predict CLV for customers based on transaction data. Data comes from two different sources; e-commerce and retail. Two data cover different customers and the date range. E-commerce data only includes Salewa data, while retail data includes data from other brands as well. For this, two different data must be standardised and any incompatible or missing data must be handled. In addition, Salewa transactions in the retail data must be filtered. After both data are ready and processable, they should be concatenated appropriately. Then, the concatenated data should be made suitable for CLV calculation. The formula for CLV calculation and the models that will make predictions should be determined. The accuracy of the predictions made by the models should be measured with different metrics and the results should be analyzed with different aspects at the same time.

Proposed Solution

First of all, we took CODEOFARTICLE numbers starting with "00-" to filter only Salewa transactions in Retail data. Since there was almost no missing value in the data, we removed very few null records.

To ensure standardization, we collected DATE data in different time formats in both data into two different columns as DATE and TIME in both data.

As explained in the following sections, certain specific columns were selected from the two data and two data were concatenated on these columns. With concatenated data, first of all, REVENUE column was created for each customer with QUANTITY*PRICE calculation.

Then, to obtain the variables to be used in CLV calculation, the necessary frequency, recency, tenure and monetary value data were obtained using the lifetimes library.

As a result of the different research we made, we decided to use this CLV formula:

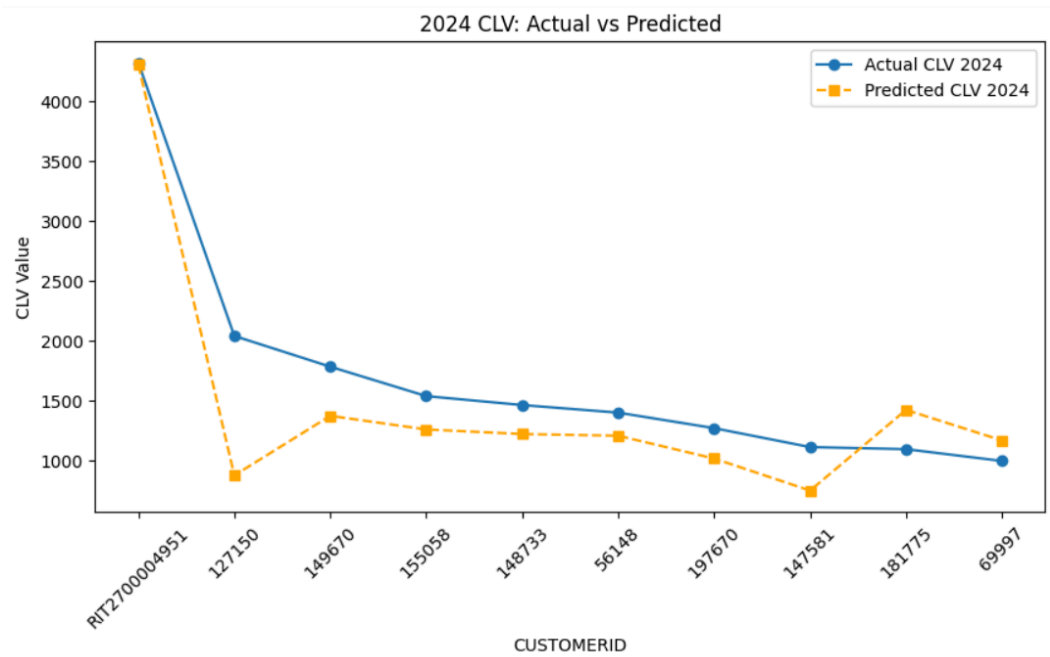
*CLV = MONETARY VALUE * FREQUENCY / (FREQUENCY + 1)*.

With this formula, the CLV value of each customer for the year 2024 was calculated.

Later, using historical data, different models such as XGBoost, ExtraTreeRegressor and MLP were used with feature engineering to estimate CLV. The predictions made by the models using the formula were compared with the different metrics we mentioned in the following sections.

CLV and accuracy results were interpreted in different ways and a future estimate was made for 2025 with the additional findings.

Summary of Findings



The graph shows that the model(XGBoost) provided the closest estimates to actual CLV values but with notable deviations for customers with exceptionally high values.

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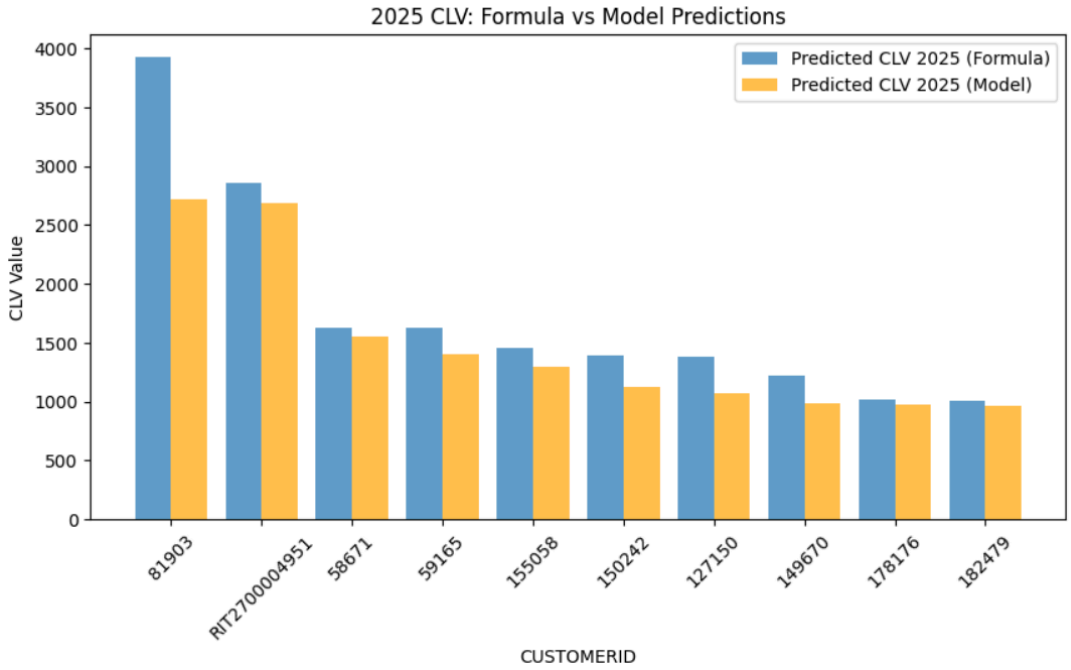
XGBoost {'MAE': 0.029565751303147394, 'RMSE': 0.04825162263335178
Extra Trees Regressor {'MAE': 0.07541117531896653, 'RMSE': 0.1318725272167586
MLPRegressor (Neural Network - Sklearn) {'MAE': 0.04665002040821227, 'RMSE': 0.07369196668045848

Accuracy Metrics for Actual vs. Predicted CLV 2024
◆ MAPE: 2.14%
◆ R² Score: 0.9658
```

- **XGBoost** achieved the highest R^2 score (%96.58), indicating that it explained a significant portion of the variance in CLV values.
- **ETR** performed moderately well but with slightly lower accuracy than XGB.
- **MLP** had the weakest performance, likely due to insufficient data complexity capture.

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XGBoost {'MAE': 6.756654103048127, 'RMSE': 39.404656729324415, 'R2': 0.8700328091187473}
```
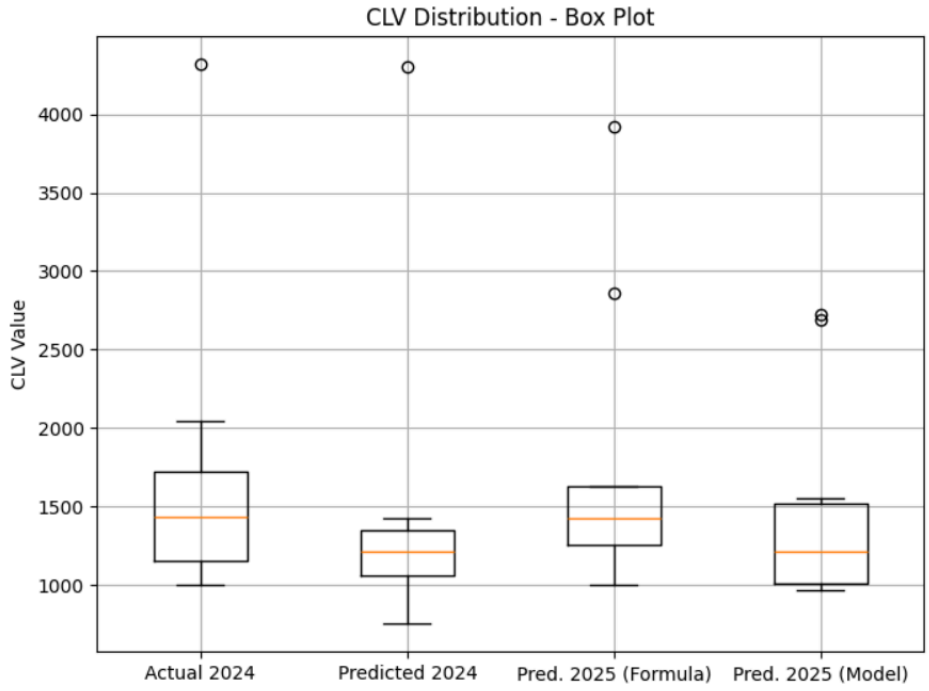

- Since the data used for the 2025 prediction came from both historical data and 2024, the model's prediction accuracy decreased compared to 2024 and became 87%.
- The formula-based method predicted higher CLV values on average.
- The model-based predictions were slightly more conservative, possibly adjusting for past variances.
- Some customers who were in the **2024 CLV top list were not present in the 2025 forecast**, indicating a shift in purchase behaviour.



- Actual 2024 CLV has the highest median but also exhibits more variability, with some extreme outliers.
- Predicted 2024 CLV is more stable, with a slightly lower median and a narrower spread.
- Formula-based 2025 predictions are slightly higher than the model-based ones, suggesting a different growth assumption.
- Outliers are present across all estimates, indicating a few high-value customers with significantly higher CLV.

Data Sources

To analyse CLV across e-commerce and retail transactions of the Oberalp Group, specifically focusing on the Salewa brand, we utilised two primary datasets: one from online transactions

(e-commerce) and another from store transactions (retail). Below are the specifics and preprocessing steps undertaken for each dataset:

E-commerce Transactions

- Source: Only transactions from Salewa's online shop are included.
- Key Fields:
    - id: Identity column unique to each transaction.
    - ordernumber and ordertime: Unique order identifier and timestamp of the order.
    - customernumber and oberalp_id: Customer identifiers, with oberalp_id linking to CRM for omnichannel tracking.
    - articleordernumber: SKU ordered, filtered to include only those starting with '00_' to focus on specific Salewa products.
    - price and quantity: Unit price and quantity ordered, used to calculate revenue as price * quantity.
    - rma_article and rma_quantity: Data on returned items, excluded from sales calculations.

Retail Transactions

- Source: Includes transactions from various stores (Salewa, Dynafit, Mountain Shop, Under Armour) but filtered to retain only Salewa-related sales.
- Key Fields:
    - Date and time: Timestamps converted to datetime format to standardize the data.

- CODEARTICLE: SKU order number, similarly filtered to include entries starting with '00_' to focus on Salewa products.
- totaltaxincluded and quantity: Total price including tax (only positive values considered to ensure accurate revenue calculations) and quantity sold, with revenue calculated as total tax included * quantity.
- oberalpid: Customer ID linked to CRM for tracking cross-channel customer activity.

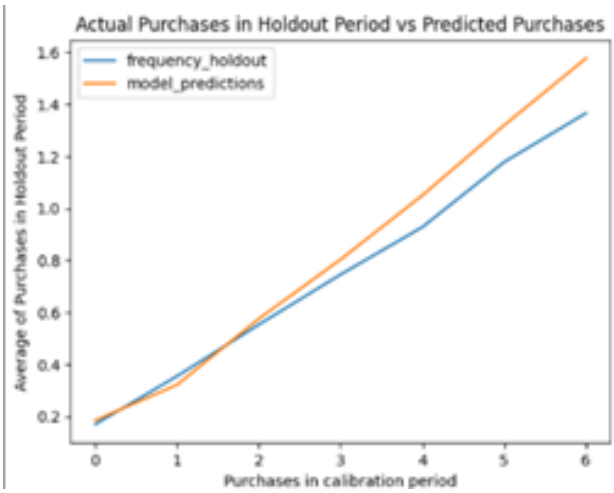
Data Preprocessing

- Datetime Conversion: Both datasets required the standardization of datetime information to ensure consistency in temporal analysis to understand and work better with frequency parameter.
- Filtering Positive Transactions: For retail data, transactions were filtered to exclude returns (negative values and zero values in the totaltaxincluded column).
- Revenue Calculation: A new column for revenue was created for both datasets based on the product of price and quantity for e-commerce, and total tax included and quantity for retail, revenue column is important for finding monetary value. Monetary value helps to predict future CLV.
- Product Filtering: Here we are only keeping those products/transactions that contain ‘00_’ in retail and e-commerce because they represent the Salewa transactions that are an important part of this project.

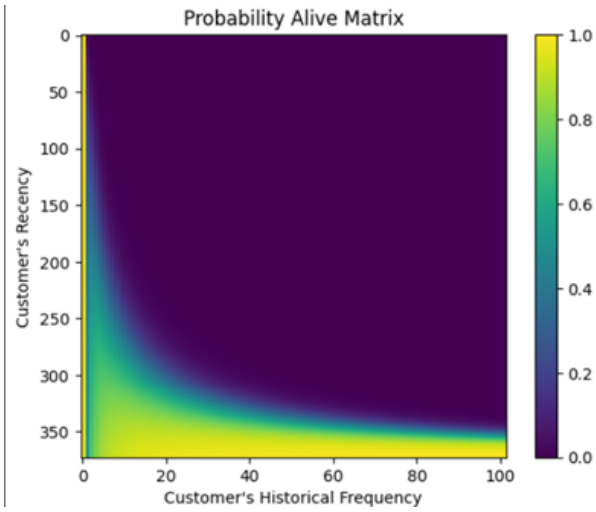
Solution and Software Architecture Description

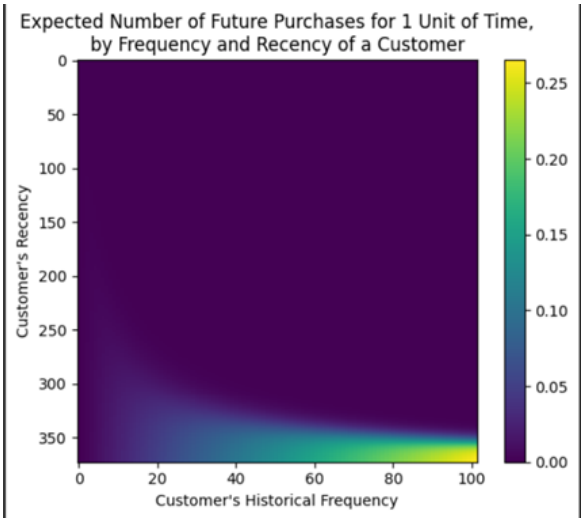
The solution to predicting CLV is structured into distinct phases, ensuring that the analysis and modelling process captures the most relevant factors influencing customer behaviour. The logical approach includes data preprocessing, as we mentioned about it in the previous sections.

Then transaction data was transformed using the lifetimes library in accordance with the CLV calculation.



Before calculating CLV, we performed the data discovery that we presented previously. Below are some graphs from our previous presentation.





We have already mentioned our formula for CLV calculation: $$CLV = \frac{\text{Monetary Value} \times \text{Frequency}}{\text{Frequency} + 1}$$ This serves as a baseline for comparing ML models. For ML-based prediction, we used feature engineering. Additional variables are engineered to improve predictive performance, such as:

- Customer lifespan: This metric represents the time between a customer's first and most recent purchase. It helps to determine how long a customer remains active in the system. Longer lifespans indicate more loyal customers, which is critical for predicting future spending behaviour in CLV models.
- Average time between purchases: This feature calculates the average interval between a customer's transactions. It helps understand a customer's purchase patterns whether they buy frequently or sporadically.
- Purchase frequency ratio: This ratio is derived by dividing the number of purchases a customer has made by the total duration of their activity (tenure). A higher purchase frequency ratio suggests a more engaged and valuable customer
- Log transformations for monetary values: Monetary values generally have a highly skewed distribution, applying a log transformation helps us to normalize them. This makes it easier for ML models to learn from the data, reducing the impact of extreme outliers and improving predictive accuracy.

Then we used feature scaling and normalization. It is essential preprocessing steps that we ensure all numerical variables are within a comparable range, improving model performance. Standard

scaling is used for transforming variables to have a mean of 0 and a standard deviation of 1, which is useful for models that rely on gradient descent. With Quantile Transformation (Normal Distribution), we ensure that numerical features follow a normal distribution, helping models to perform better. These scaling techniques helped us to avoid biases toward large numerical values and prevent any particular feature from dominating the learning process.

Predicting CLV is complex because customer purchasing behaviour is most of the time non-linear, meaning traditional linear models struggle to capture patterns. This is why we think tree-based and deep-learning models are crucial.

XGBoost: This gradient-boosting algorithm efficiently identifies non-linear relationships in customer behaviour. It assigns higher weights to critical customer segments, allowing it to prioritize loyal customers in CLV prediction. Its ability to rank feature importance helps businesses understand which factors drive customer lifetime value the most.

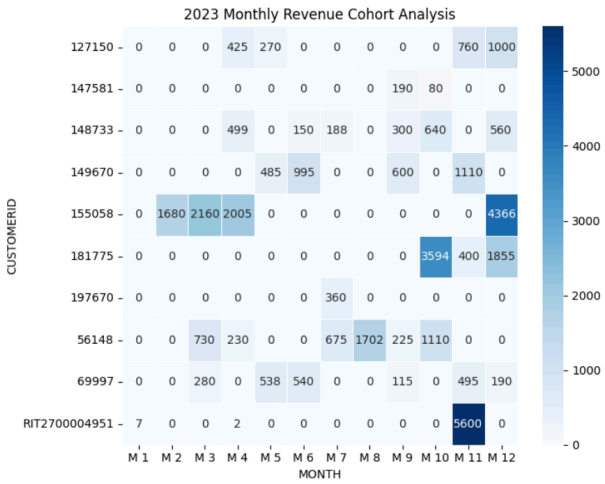
Extra Trees Regressor: This ensemble learning technique introduces randomness into decision trees, reducing the likelihood of overfitting. It is particularly effective in capturing the relationship between purchase frequency and monetary value. By leveraging multiple randomized decision trees, ETR can generalize well to unseen customers.

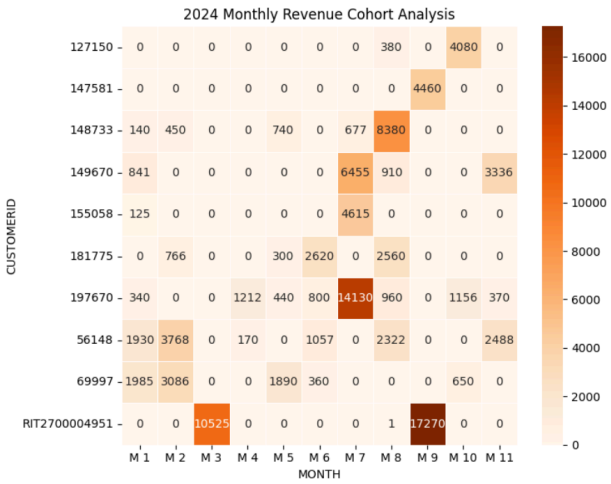
Multi-Layer Perceptron: Unlike decision trees, MLP can learn deep, hidden patterns in customer purchase behaviour. It is particularly useful when analyzing long-term customer engagement trends, making it valuable for predicting future spending. MLP performs well when CLV estimation involves sequential or time-dependent purchasing behaviours.

BG/NBD model is used for probabilistic CLV estimation based on purchase frequency and recency. Even though we did not use it in CLV prediction, our experiments with this model helped us in our later experiments with other models.

As stated in the Summary of Findings section, we measured the accuracy of the models with MAE, RMSE, and R^2 metrics and found that XGBoost achieved the highest accuracy. However, we compared the CLVs calculated by the formula with the CLVs predicted by the model. We explain the problems we encountered in the following sections. 2024 CLV was calculated using both 2023 data with the same formula and model. Also, the average of 2023 and 2024 data was used for the 2025 future prediction. Separate top 10 lists for these years were determined according to CLV.

Then, a monthly cohort analysis of the top 10 customers with the highest CLV of 2024 was conducted for 2023 and 2024.





The above cohort tables were analyzed and the purchasing habits of each customer individually were interpreted in the ipynb file.

Solution Evaluation and Conclusions

The evaluation phase of this CLV prediction project focuses on assessing model performance, identifying challenges in the estimation process, and analyzing how different methodologies affect prediction accuracy.

XGBoost outperformed the other models in terms of generalization and predictive power. ETR and MLP models performed well in specific scenarios but showed limitations in handling outliers and extreme values. BG/NBD Model Accuracy was insufficient, reaching only 63%, failing to generalize CLV trends effectively.

CLV estimation is inherently complex and non-linear, making it challenging to develop a model that generalizes well across all customers. Several key challenges were identified during the evaluation:

- Customers exhibit significant differences in purchase behaviour across the years. Some customers make frequent purchases, while others have sporadic buying patterns, making generalization difficult. The presence of outliers and customers with near-zero CLV values complicates model training.

- Initial experiments showed that using all data for model training led to lower accuracy due to the high variability across years. To address this, we opted to train the models using only 2023 data, as it provided higher accuracy and more consistent results.
- The shorter the observation period and the closer the data is to the present, the higher the accuracy of predictions. Predictions for 2024 were more accurate than those for 2025. This is because the 2025 forecast relied on more historical data, making it harder to generalize future trends.
- Several CLV formulas exist, each with different assumptions and applications. The formula used in this project was chosen due to its alignment with B2C business models and prior successful implementations in similar projects. This formula has been shown to yield realistic and reliable CLV estimates based on previous research.

To enhance the model, several improvements can be considered in the future:

- Implementing ARIMA or Prophet models could improve long-term forecasting accuracy.
- Using LSTM or Transformer-based models may help capture sequential customer behaviours.
- Combining BG/NBD with machine learning models could enhance purchase probability predictions.

- Incorporating customer segmentation may lead to more tailored predictions.

This project demonstrates that machine learning significantly enhances CLV estimation compared to traditional methods. The findings provide practical insights for businesses seeking to optimize customer retention, revenue forecasting, and strategic planning.

By continuously improving data selection, model architectures, and feature engineering, CLV predictions can be made even more accurate and actionable, leading to better decision-making in customer management and marketing strategies.